Chapter 21 & 13: Consumer theory & Costs of production

Rudy Kretschmar
Discussion section 1

March 11, 2026

Outline

Today we will talk about:

- The theory of the consumer
  - budget constraints
  - indifference curves
- Costs of production

Budget constraints

Most consumers want to increase the quantity/quality of the goods they consume, but money isn't infinite :(

Consumers face a **budget constraint:**

the limit on the consumption bundles that a consumer can afford

Example: A consumer would like to purchase pasta and wine. They have $100 to spend, pasta costs $20 per plate and wine costs $10 per glass...

1. How much of each good could they consume?
2. What is the equation for their budget constraint?
3. What does their budget constraint look like graphically?
4. What happens if the price of pasta decreases to $10?
5. What happens if income increases to $160?

Representing preferences

How do we know which bundle of goods a consumer prefers?

We can represent preferences using **indifference curves:**

IC: a curve showing consumption bundles that give the consumer *equal levels of satisfaction*

The slope at any point on an indifference curve equals the **Marginal Rate of Substitution (MRS)**: the rate at which a consumer is willing to give up one unit of good X for another unit of good Y

We calculate MRS as follows:

$$MRS = \frac{MU_X}{MU_Y}$$

Properties of indifference curves:

1. Higher indifference curves are preferred to lower ones:
   1. More is better!
2. Indifference curves are downward sloping:
   1. in most cases a consumer likes both goods, so to decrease plates of pasta but keep the same satisfaction, we must increase glasses of wine
3. Indifference curves do not cross
   1. If they crossed, we would violate the assumption that people prefer more to less
4. Indifference curves are bowed inwards
   1. MRS is not constant... we are more willing to give up a good we already have an abundance of

Optimizing

A consumer wants to maximize their satisfaction given their budget constraint...

This is achieved by choosing the bundle where either:

- Their indifference curve is tangent to their budget constraint (where they touch... kind of)
- The MRS is equal to the slope of the budget constraint

When income or prices of goods change, our budget constraint does too, and so we would find ourselves on a new indifference curve at optimum...

Income & Substitution Effects

When the price of goods changes, the resulting change to the budget constraint and optimal bundle is a result of two effects:

1. **Income effect:** Change in consumption that results when a price change *moves the consumer to a higher or lower indifference curve*
2. **Substitution effect:** Change in consumption that results when a price change *moves the consumer along a given indifference curve to a point with a new MRS*

** A change in income results only in an income effect **

Putting It All Together

Let's Draw an Example where we experience a price change of one good.

1. According to the income effect, will we demand more or less of each good?
2. According to the substitution effect, will we demand more or less of each good?
3. What can we say happens to the consumption bundle? Is anything left ambiguous?

Further Discussion

Consumer theory is the primitive from which we back out the Law of Demand

Your textbook gives a good treatment on a few things that I think are worth pointing out. These are:

1. Giffen Goods (Show that indifference curves of a certain shape imply a good is a Giffen Good)
2. Wages and the labor supply (See if you are interested in microeconomics or macroeconomics research)
3. Interest Rates and Household Savings
   - This treat consumption today and consumption tomorrow as two different goods. You can think of the interest rate being a change in the relative price of the two.
   - This concept is the foundational for most of the macroeconomic theory you see at the grad school level. See discussion in the book if you are interested in macro or finance.

Costs of Production

A firm aims to maximize their profit:

$$\text{Profit} = \text{total revenue} - \text{total cost}$$

- total revenue: $TR = P \times Q$
- total cost is the sum of all **implicit** and **explicit** costs of inputs to production
  - **Implicit**: input costs which *do not* require monetary outlay from the firm
  - **Explicit**: input costs which *do* require a monetary outlay from the firm

Types of Profit

There are two types of profit we can consider:

Economic Profit:

total revenue less implicit *and* explicit costs

Accounting

total revenue less explicit costs *only*

Costs

Types of costs:

- Fixed costs: costs that do not vary with Q
- Variable costs: costs that vary with Q

Measures of costs:

- Average total cost (ATC): $ATC = TC/Q$
- Average fixed cost (AFC): $AFC = FC/Q$
- Average variable cost (AVC): $AVC = VC/Q$
- Marginal Cost: $MC = \Delta TC/\Delta Q$

Example

Consider a firm which produces pizzas:

A worker costs \$100 and the firm has fixed costs of \$200.

The following table shows the number of pizzas produced given varying numbers of workers. Let's add information for the marginal product, total cost, average total cost, and marginal cost.

Workers	Output
0	0
1	20
2	45
3	80
4	100
5	110

What do costs look like?

- We tend to assume (like in previous example) that the marginal product of labor is decreasing... Why?
- Average fixed cost falls as output rises because the same fixed cost is spread over more units.
- Average variable cost typically falls at first and then rises: initially, workers can specialize and use fixed inputs more effectively, but eventually diminishing marginal product sets in.
- Marginal cost also often has a U-shape because when marginal product is rising, each additional unit is relatively cheap to produce, but when marginal product falls, each additional unit becomes more costly.
- Average total cost is the sum of average fixed cost and average variable cost, so it often has a U-shape: it falls at first, then rises once the increase in AVC outweighs the continued decline in AFC.

What do costs look like?

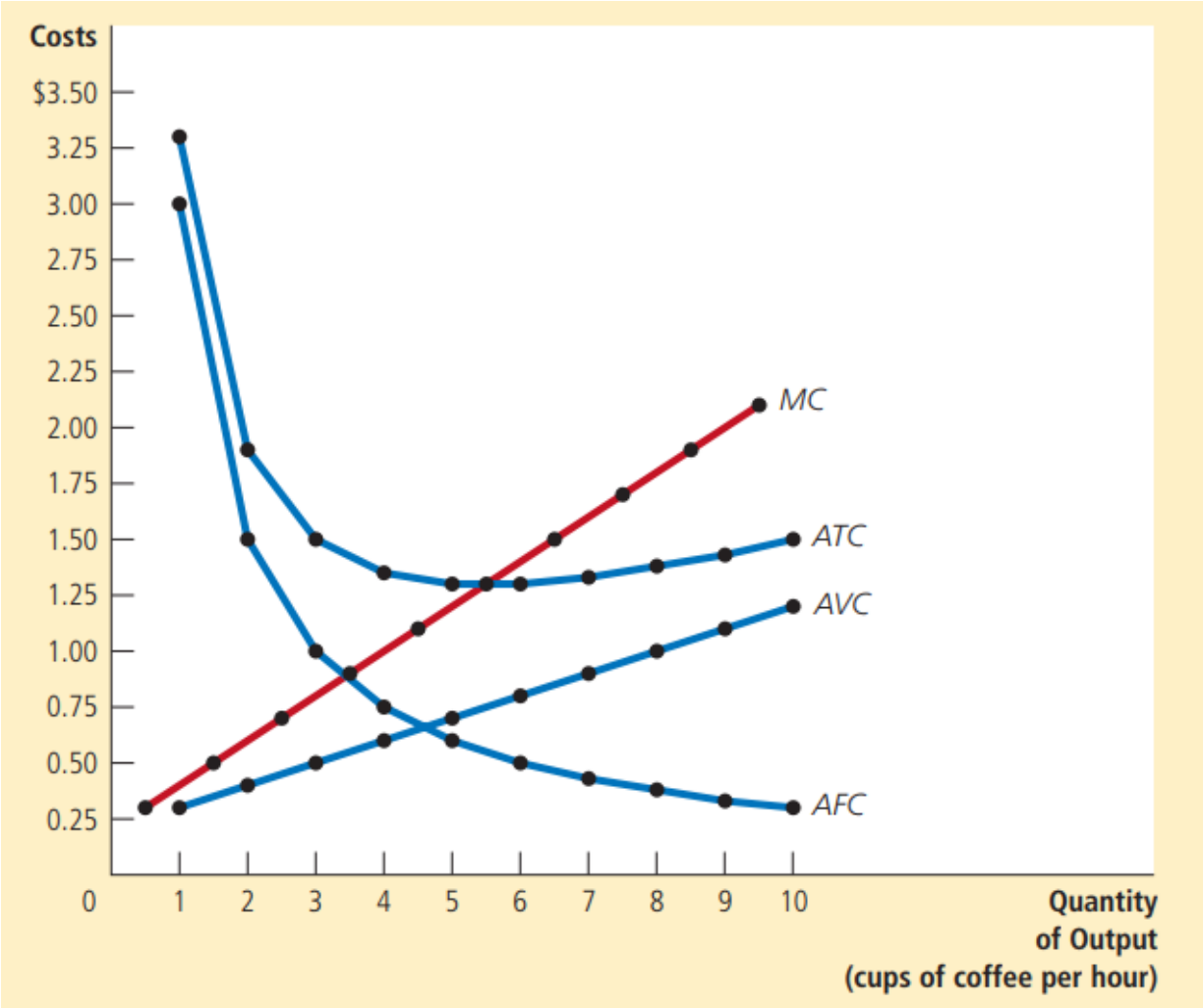

Figure 1: Relationship between costs

A Final Piece of Intuition

The marginal cost curve intersects average total cost at the minimum of ATC: if MC is below ATC, producing one more unit pulls the average down; if MC is above ATC, producing one more unit pushes the average up.

Economies and Diseconomies of Scale

We can define the scale of production in the following 3 ways:

1. Economies of Scale: Long run ATC declines as Q increases
2. Diseconomies of Scale: Long run ATC increases as Q increases
3. Constant returns to scale: Long run ATC stays the same as Q changes